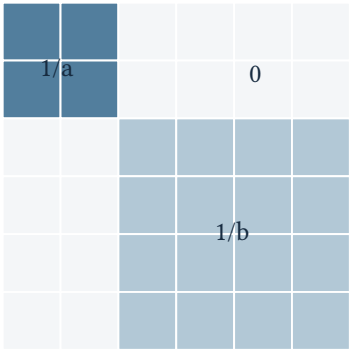
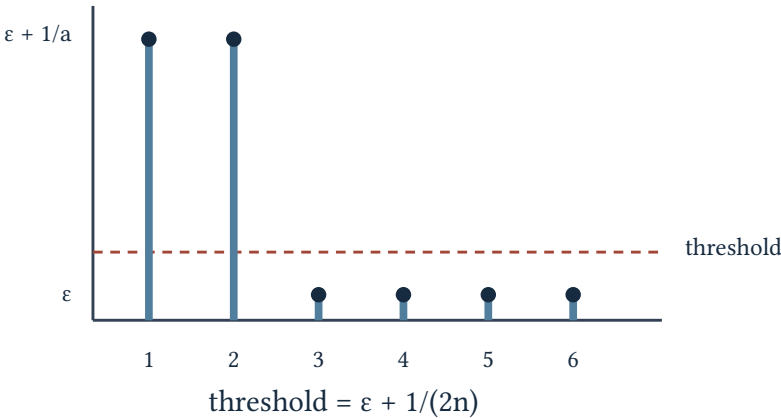


Two-block averaging matrix

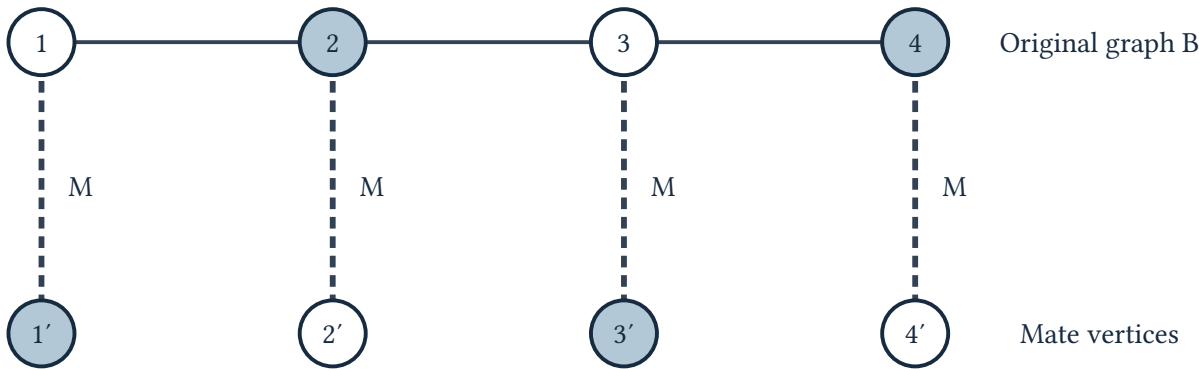


$P_s : a = 2, b = 4$

A row separates the two groups

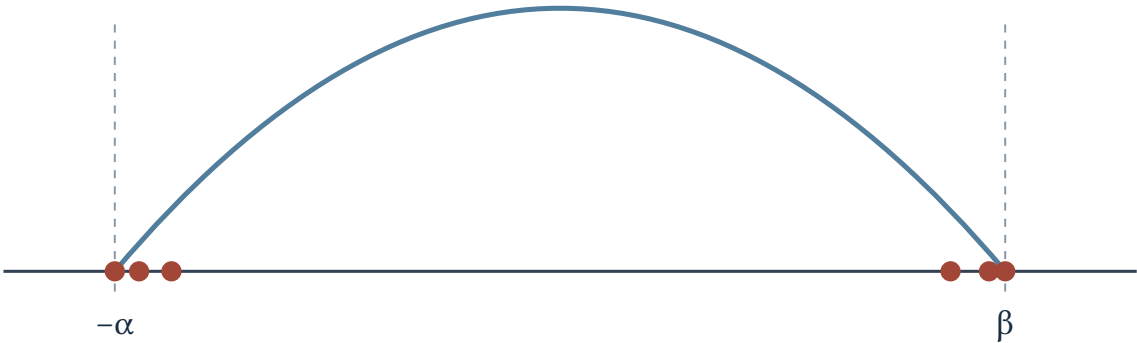


Heavy matching enforces balanced cuts and separated mates



Every accepted density cut crosses all matching edges; complementing weights reverses the objective.

Nonnegativity confines a near-optimal direction to two levels



$$\sum (v_i + \alpha)(\beta - v_i) = n\alpha\beta - 1$$

A small nonnegative sum forces every coordinate close to an endpoint.